Safety of MRAC + min-norm CBF augmentation with an unknown A matrix: is the tube necessary?

Notes for `Robust_Adaptive.m` and `Robust_Adaptive_TK.m` (lateral-directional aircraft example, Menner & Lavretsky). October 2026; revised October 21 to match the current `Robust_Adaptive_TK.m`.

Short answers.

1. **Your reasoning is correct.** While the gains are adapting, the aircraft does not behave like any model the controller can compute. A barrier condition evaluated without a margin therefore cannot guarantee the original bounds. I had said the tube was not needed for the earlier version of `Robust_Adaptive_TK.m`, which ran the CBF on the plant. That was too strong: that design gives a *bounded and vanishing* violation (Theorem 1), not a guarantee. The β peak of 0.5005° it produced is exactly such a violation. The current `Robust_Adaptive_TK.m` implements the tube (Theorem 2).
2. **Your tube gives a hard guarantee** (Theorem 2), with one refinement to the picture. The CBF acts on the *reference model*, which is known exactly, so the reference model never violates the shrunken bounds. The *plant* may cross the shrunken bounds by up to $\delta_i(t)$, the full tube width rather than just "minimally", but it can never cross the original bounds.
3. **The guarantee needs four things:** a known bound $\theta_{\max}$ on the uncertainty; a non-empty shrunken box (δ_i smaller than the half-width of the bounds); $x(0) = x_{\mathrm{ref}}(0)$; and measurement of the full state, so that e is known (assumption (A6)). Clamping the tube, as in `min(delta, 0.9*Y_max)`, voids the guarantee.
4. **Why your tube looked unusable.** With $\gamma = 10^4$ and the open-loop reference model, the tube at $t = 0$ is 3.99° for the rudder (limit 2°) and 0.78° for β (limit 0.5°). The shrunken box was empty. The tube width scales like $\theta_{\max}/\sqrt{\gamma}$. With the observer-like reference model, a larger γ is affordable: at $\gamma = 10^5$ the tube is $[0.29^\circ,\ 0.65^\circ,\ 0.27\,\mathrm{dps},\ 0.26^\circ]$, which is feasible. The current `Robust_Adaptive_TK.m` uses the open-loop reference model with $\gamma = 5 \cdot 10^4$: tube $[0.49^\circ,\ 1.78^\circ,\ 1.17\,\mathrm{dps},\ 0.35^\circ]$, also feasible (Section 6).
5. **Safety can be guaranteed without shrinking the bounds** by putting the margin on the barrier's *derivative* instead (Proposition 3, a robust CBF). Some margin, however, is unavoidable (Remark 4). For this aircraft the derivative-margin version is infeasible on the rudder row, so your tube is the right tool here.
6. **The tube guarantees u_{bl}, p_s and β, not the total command $u_{\mathrm{bl}} + w$.** In one run the tube case kept u_{bl} inside its limits while the total rudder command reached $1.33\times$ its limit (Table 2). Without the tube, the violation of the guaranteed quantities is capped by the tube width, so it stays small whenever the tube would fit (Corollary 2).

1. Setup

The extended state is $x = [e_{yI}^\top\ x_p^\top]^\top \in \mathbb{R}^n$ ($n = 5, m = 2$). The true plant has an unknown matrix $\Theta \in \mathbb{R}^{m\times n}$ that enters through the known input matrix B (matched uncertainty). In the code this is `A_true = A + B*Theta`:

$$\dot{x} = (A + B\Theta)\,x + B\,u + B_{\mathrm{ref}}\,y_{\mathrm{cmd}} + B_v\,v, \qquad B_{\mathrm{ref}} = -B_v.$$

A, B, B_v are known. The baseline LQR gain gives the Hurwitz matrix $A_{\mathrm{ref}} = A - BK_{\mathrm{lqr}}$, and the *ideal* gain $K^\star = K_{\mathrm{lqr}} + \Theta$ satisfies $A + B\Theta - BK^\star = A_{\mathrm{ref}}$. The controller and the reference model are

$$u = -\hat{K}x + w, \qquad \dot{x}_{\mathrm{ref}} = A_{\mathrm{ref}}x_{\mathrm{ref}} + B_\pi(\pi_0 + \pi) + L_v\,(x - x_{\mathrm{ref}}),$$

with $B_\pi = [B_v \;\; B]$, $\pi = [v^\top \; w^\top]^\top$ the CBF signals, and $\pi_0 = [-y_{\text{cmd}}^\top \; 0]^\top$, so that $B_\pi \pi_0 = B_{\text{ref}} y_{\text{cmd}}$. Both the original file and the current `Robust_Adaptive_TK.m` use $L_v = 0$ (open-loop reference model). The observer-like reference model $L_v = P_v R_v^{-1}$, used by an earlier version of the TK file, is kept there as a commented option and is also covered. Define $e = x - x_{\text{ref}}$, $\Delta K = \hat{K} - K^\star$ and $A_e = A_{\text{ref}} - L_v$ (Hurwitz). Let $P = P^\top \succ 0$ satisfy $A_e^\top P + P A_e = -\tilde{Q}$ with $\tilde{Q} \succ 0$. With $L_v = 0$ the code uses $P =$ `lyap(A_ref',I)`, so $\tilde{Q} = I$. With the observer-like model it uses $P = P_v^{-1}$. The code computes $\tilde{Q}$ from whichever pair it uses (`Q_tilde`). The adaptive law, with $\Gamma = \text{blkdiag}(\Gamma_I, \Gamma_P) \succ 0$, is

$$\dot{\hat{K}} = B^\top P \, e \, x^\top \Gamma.$$

Constrained outputs. $y_{\text{lim}} = C_{\text{lim}}(t)\, x$ with $C_{\text{lim}}(t) = \begin{bmatrix} -\hat{K}(t) \\ 0 \;\; C_{p\,\text{lim}} \end{bmatrix}$. The rows $c_i(t)$, $i = 1, \ldots, 2m$, are: $u_{\text{bl}} = -\hat{K} x$ (aileron, rudder), then p_s and β. Bounds hold componentwise, $y^{\min} \le y_{\text{lim}} \le y^{\max}$ with $y_i^{\min} < y_i^{\max}$. All rows are treated as relative degree one with slopes $\alpha = \text{diag}(\alpha_i) \succ 0$, and

$$H_x = C_{\text{lim}} A_{\text{ref}} + \alpha C_{\text{lim}}, \qquad H_u = C_{\text{lim}} B_\pi = \begin{bmatrix} -\hat{K}_I & -\hat{K}_I D_{p\,\text{reg}} - \hat{K}_P B_p \\ 0 & C_{p\,\text{lim}} B_p \end{bmatrix}.$$

Where the barrier condition $\dot{h} + \alpha h \le 0$ comes from

Each limit is written as a barrier $h_i \le 0$. For the max side, $h_i = c_i x - y_i^{\max}$, the negative of the distance to the limit. The CBF signals π do not appear in h_i itself, so $h_i \le 0$ cannot be imposed on π directly. They do appear in $\dot{h}_i$, because every row has relative degree one. The design therefore imposes a condition on $\dot{h}_i$, chosen so that it implies $h_i \le 0$ for all time.

Class-$\mathcal{K}$ functions. A continuous function $\kappa : [0, a) \to [0, \infty)$ is *class-$\mathcal{K}$* if it is strictly increasing and $\kappa(0) = 0$. It is *extended class-$\mathcal{K}$* if it is defined on all of $\mathbb{R}$, continuous, strictly increasing, and $\kappa(0) = 0$, so it is negative for negative arguments. In the CBF literature the safe set is usually written $\{h_s \ge 0\}$, and the CBF condition is $\dot{h}_s \ge -\kappa(h_s)$ for an extended class-$\mathcal{K}$ function κ [Ames et al. 2017]. With our sign convention, $h = -h_s$, this becomes

$$\dot{h} \;\le\; \kappa(-h).$$

This is a speed limit that depends on the distance to the boundary:

- **Inside the box** ($h < 0$), $\kappa(-h) > 0$: the quantity may approach its limit, at a speed of at most $\kappa(\text{distance})$.
- **On the boundary** ($h = 0$), the condition reduces to $\dot{h} \le \kappa(0) = 0$. This is Nagumo's condition: the flow may not point outward [Nagumo 1942; Blanchini 1999].

The linear choice. Both papers use $\kappa(r) = \alpha_i r$ with $\alpha_i > 0$, which gives

$$\dot{h}_i \le -\alpha_i h_i \quad \Longleftrightarrow \quad \dot{h}_i + \alpha_i h_i \le 0.$$

In their notation, this is the stable polynomial $\varphi_i(s) = s + \alpha_i$ applied to h_i as a derivative operator, $\varphi_i(s) h_i = \dot{h}_i + \alpha_i h_i$. For relative degree r, φ_i has r stable roots, which is the exponential CBF of [Nguyen & Sreenath 2016]. So $\alpha_i h_i$ is not a damping term. It is the class-$\mathcal{K}$ term evaluated at the distance to the limit, and it caps the approach rate with time constant $1/\alpha_i$. The linear choice is what keeps the condition linear in both x and π. That is why H_x contains the term αC_{lim}, and why the closed-form CBF is piecewise linear. A nonlinear κ would also be valid, but it would lose that structure.

Fact 1 (comparison lemma; Khalil 2002, Lemma 3.4). Let $f(t,u)$ be continuous in t and locally Lipschitz in u, and let $u(t)$ solve $\dot{u} = f(t,u), u(0) = u_0$. If $v(t)$ is continuous, its upper right derivative satisfies $D^+v(t) \le f(t,v(t))$, and $v(0) \le u_0$, then $v(t) \le u(t)$ for all t in the interval of existence.

Fact 2 (the barrier condition keeps the box invariant). Let κ be a locally Lipschitz extended class-$\mathcal{K}$ function. If $\dot{h}(t) \le \kappa(-h(t))$ for all $t \ge 0$ and $h(0) \le 0$, then $h(t) \le 0$ for all $t \ge 0$. For $\kappa(r) = \alpha r$, more precisely, $h(t) \le e^{-\alpha t}h(0)$.

Proof. Compare h with the solution y of $\dot{y} = \kappa(-y), y(0) = h(0)$. Since $y \equiv 0$ is an equilibrium and solutions are unique, $y(0) \le 0$ implies $y(t) \le 0$ for all t. Fact 1 then gives $h(t) \le y(t) \le 0$. For $\kappa(r) = \alpha r$, the comparison solution is $y(t) = e^{-\alpha t}h(0)$. ■

This is the forward-invariance result for zeroing CBFs [Ames et al. 2017], stated for one barrier along a given trajectory. Lavretsky & Menner's Lemma 2 [ACC 2025] is the same argument for a general relative degree. Fact 2 is what Theorems 1 and 2 below use. In Theorem 1 the right-hand side also contains the unknown model error d_i. In Theorem 2 the barrier contains the time-varying tube $\delta_i(t)$, which is why $\dot{\delta}_i$ appears.

Assumptions

(A1) Matching. The uncertainty enters as $B\Theta$, with B known.

(A2) Prior bound. $\|K^\star - \hat{K}(0)\|_F = \|\Theta\|_F \le \theta_{\max}$, with $\theta_{\max}$ known. Code: $\hat{K}(0) = K_{\text{lqr}}$ and $\theta_{\max} = 1.5$; the true value is 1.3625.

(A3) Same initial state. $x(0) = x_{\text{ref}}(0)$, so $e(0) = 0$.

(A4) Invertible H_u. $H_u(t)$ is invertible for all t, i.e. $\hat{K}_I(t)$ and $C_{p\,\text{lim}}B_p$ are invertible.

(A5) Existence of solutions. Solutions exist on $[0,\infty)$. The closed loop is locally Lipschitz because the CBF law is continuous and piecewise linear.

(A6) Full-state measurement. The whole extended state $x = [e_{yI}^\top \; x_p^\top]^\top$ is measured, including β (both papers' Assumption 1). Together with x_{ref}, which the controller computes, this makes $e = x - x_{\text{ref}}$ known. The adaptive law, $\bar{V}(t)$, $\delta_i(t)$ and the CBF signals are then all computable. If a state such as β is estimated instead, as in `OBLTR.m`, the controller only knows $\hat{x} - x_{\text{ref}}$, and the estimation error would have to be added to the tube. That case is not covered here.

2. Three lemmas

Lemma 1 (closed-form CBF satisfies every modified constraint). Let $a, b \in \mathbb{R}^{2m}$ with $a_i + b_i < 0$ for every i, and $\pi = H_u^{-1}\big(\max(0,b) - \max(0,a)\big)$. Then $a + H_u\pi \le 0$ and $b - H_u\pi \le 0$ componentwise.

Proof. $(H_u\pi)_i = \max(0,b_i) - \max(0,a_i)$. If $a_i > 0$, then $b_i < -a_i < 0$, so $(H_u\pi)_i = -a_i$, $a_i + (H_u\pi)_i = 0$ and $b_i - (H_u\pi)_i = a_i + b_i < 0$. The case $b_i > 0$ is symmetric. If $a_i, b_i \le 0$, then $(H_u\pi)_i = 0$. ■

In the code, `H_max_delta` plays the role of a and `H_min_delta` of b. Without tightening, $a_i + b_i = -\alpha_i(y_i^{\max} - y_i^{\min}) < 0$ always holds. With tightening, this becomes the feasibility condition of Theorem 2.

Lemma 2 (certainty-equivalence identity). For any $\hat{K}$, the estimated plant $\hat{A} = A + B(\hat{K} - K_{\text{lqr}})$ satisfies $\hat{A} - B\hat{K} = A_{\text{ref}}$. The true closed loop satisfies $A + B\Theta - B\hat{K} = A_{\text{ref}} - B\,\Delta K$.

Proof. Direct substitution, using $\Theta = K^\star - K_{\text{lqr}}$. ■

This is why H_x is built from A_{ref} even though A is unknown. The only unknown left in the plant's barrier derivative is $-c_i B\,\Delta K\,x$.

Lemma 3 (MRAC Lyapunov function and the tube bound). Let $V = e^\top P e + \operatorname{tr}(\Delta K\,\Gamma^{-1}\Delta K^\top)$ and define the computable, non-increasing signal

$$\bar{V}(t) = \bar{V}_0 - \int_0^t e^\top \tilde{Q}\, e\, d\tau, \qquad \bar{V}_0 = \theta_{\max}^2\,\lambda_{\max}(\Gamma^{-1}) \quad (= \theta_{\max}^2/\gamma \text{ for } \Gamma = \gamma I).$$

Then, under (A1)–(A3) and (A6), $0 \le V(t) \le \bar{V}(t)$ for all $t \ge 0$, and for every row vector c:

$$\text{(a)}\ \ |c\,e(t)| \le \sqrt{c\,P^{-1}c^\top}\,\sqrt{\bar{V}(t)}, \qquad \text{(b)}\ \ \|\Delta K(t)\|_F^2 \le \lambda_{\max}(\Gamma)\,\bar{V}(t).$$

Proof. By Lemma 2, $\dot{x} = (A_{\text{ref}} - B\Delta K)x + B_\pi(\pi_0 + \pi)$. Subtracting the reference model gives

$$\dot{e} = A_e\,e - B\,\Delta K\,x.$$

The CBF signals π cancel because they drive the plant and the reference model identically. The MRAC proof is therefore untouched by the CBF. Since $K^\star$ is constant, $\dot{\Delta K} = \dot{\hat{K}}$, and

$$\dot{V} = -e^\top \tilde{Q} e - 2e^\top P B \Delta K x + 2\,\operatorname{tr}\big(\Delta K \Gamma^{-1}\Gamma x e^\top P B\big) = -e^\top \tilde{Q} e,$$

because $\operatorname{tr}(\Delta K x e^\top P B) = e^\top P B \Delta K x$. Hence $V(t) = V(0) - \int_0^t e^\top \tilde{Q} e$. With $e(0) = 0$, $V(0) = \operatorname{tr}(\Delta K(0)\Gamma^{-1}\Delta K(0)^\top) \le \lambda_{\max}(\Gamma^{-1})\theta_{\max}^2 = \bar{V}_0$, so $V \le \bar{V}$. For (a), Cauchy–Schwarz in the P-norm gives $|ce| = |(cP^{-1/2})(P^{1/2}e)| \le \sqrt{cP^{-1}c^\top}\sqrt{e^\top P e} \le \sqrt{cP^{-1}c^\top}\sqrt{V}$. For (b), $\operatorname{tr}(\Delta K\Gamma^{-1}\Delta K^\top) \ge \|\Delta K\|_F^2/\lambda_{\max}(\Gamma)$. ■

Your original code implements exactly this: `V_bar0 = theta_max^2/gamma`, `int_eQe`, and `delta = sqrt(diag(C_lim*P_inv*C_lim')*V_bar)`.

Remark (what $\tilde{Q} \succ 0$ is needed for). The identity $V(t) = V(0) - \int_0^t e^\top \tilde{Q} e$ holds for *any* symmetric P used in the adaptive law, with $\tilde{Q} := -(A_e^\top P + P A_e)$. Parts (a) and (b) use only $P \succ 0$ and $V \ge e^\top P e$. So the tube bound stays valid even if $\tilde{Q}$ is indefinite, as long as $\bar{V}$ is computed with that same $\tilde{Q}$. Positive definiteness of $\tilde{Q}$ is needed for two other things: $\bar{V}$ being non-increasing, and MRAC stability (e and ΔK bounded). For example, keeping $P = P_v^{-1}$ while setting $L_v = 0$ gives $\operatorname{eig}(\tilde{Q}) = [-5.62,\ -2.06,\ 1.57,\ 10.2,\ 43.1]$. The tube is still a valid bound, but $\bar{V}$ can grow until (F) fails, and stability is no longer guaranteed. That is why the code switches P together with L_v.

3. Without a margin: CBF on the plant state (the earlier `Robust_Adaptive_TK.m`)

> **Theorem 1 (bounded, vanishing violation; no guarantee).** Suppose the CBF is evaluated on the plant state with the certainty-equivalence model:
>
> $$a = H_x x + H_u \pi_0 + \dot{C}_{\lim} x - \alpha y^{\max}, \qquad b = -H_x x - H_u \pi_0 - \dot{C}_{\lim} x + \alpha y^{\min},$$
>
> and π is given by Lemma 1. Let $d_i(t) = -c_i(t)\, B\, \Delta K(t)\, x(t)$, the model error that no measurement reveals. If $y^{\min} \le C_{\lim}(0)x(0) \le y^{\max}$, then for all $t \ge 0$
>
> $$c_i(t)x(t) \;\le\; y_i^{\max} + \int_0^t e^{-\alpha_i(t-\tau)} d_i(\tau)\, d\tau \;\le\; y_i^{\max} + \frac{1}{\alpha_i} \sup_{\tau \le t} \max\big(d_i(\tau), 0\big),$$
>
> and the min side is symmetric. The violation is bounded by the model error divided by the slope (input-to-state safety). It is zero only if $c_i B\, \Delta K\, x \equiv 0$.

Proof. Let $h_i = c_i x - y_i^{\max}$. By Lemma 2, $\dot{h}_i = \dot{c}_i x + c_i A_{\text{ref}} x + c_i B_\pi (\pi_0 + \pi) - c_i B \Delta K x$, so $\dot{h}_i + \alpha_i h_i = a_i + (H_u \pi)_i + d_i \le d_i$ by Lemma 1. Since $h_i(0) \le 0$, Fact 1 (comparison lemma) with $f(t,u) = -\alpha_i u + d_i(t)$ gives $h_i(t) \le \int_0^t e^{-\alpha_i(t-\tau)} d_i\, d\tau$. ■

Remarks. (i) If x is bounded, then $e \in L_2 \cap L_\infty$ and $\dot{e}, \ddot{e} \in L_\infty$. Barbalat's lemma then gives $e \to 0$ and $\dot{e} \to 0$, hence $B\Delta K x = A_e e - \dot{e} \to 0$ and $d_i \to 0$: the violation vanishes asymptotically. (ii) For the β row, $c_\beta B = [0 \;\; 0.0153]$, so d_β is tiny. This explains why the earlier TK version held β to 0.5005° rather than exactly 0.5°. (iii) Theorem 1 is the honest statement for that design: good performance and small violations, but no guarantee.

4. With a tube: CBF on the reference model (your idea, made rigorous; the current `Robust_Adaptive_TK.m`)

Define the tube and the shrunken bounds

$$\delta_i(t) = \sqrt{c_i(t) P^{-1} c_i(t)^\top \bar{V}(t)}, \qquad y_i^{\min} + \delta_i(t) \;\le\; c_i(t)\, x_{\text{ref}} \;\le\; y_i^{\max} - \delta_i(t).$$

The CBF is evaluated on x_{ref}. Every term of the reference model is known, including $L_v e$, which is measured. With

$$\ell_i = \big(H_x x_{\text{ref}} + H_u \pi_0 + \dot{C}_{\lim} x_{\text{ref}} + C_{\lim} L_v\, e\big)_i,$$
$$a_i = \ell_i + \dot{\delta}_i - \alpha_i\big(y_i^{\max} - \delta_i\big), \qquad b_i = -\ell_i + \dot{\delta}_i + \alpha_i\big(y_i^{\min} + \delta_i\big),$$

π is from Lemma 1, and $u = -\hat{K} x + w$.

Tube derivative $\dot{\delta}_i$. Write $\delta_i = \sqrt{s_i\, \bar{V}}$ with $s_i := c_i P^{-1} c_i^\top$. Differentiating, using that P^{-1} is symmetric and that $\dot{\bar{V}} = -e^\top \tilde{Q} e$ (Lemma 3):

$$\dot{s}_i = 2\, \dot{c}_i P^{-1} c_i^\top, \qquad \dot{\delta}_i = \frac{\dot{s}_i\, \bar{V} + s_i\, \dot{\bar{V}}}{2\, \delta_i} = \frac{2\, \dot{c}_i P^{-1} c_i^\top\, \bar{V} \;-\; c_i P^{-1} c_i^\top\, e^\top \tilde{Q} e}{2\, \delta_i}.$$

Every term is computable at each step. The row derivative differs by row:

- **Aileron and rudder rows** ($c_i = -\hat{K}_i$): $\dot{c}_i = -\dot{\hat{K}}_i$, the i-th row of $-B^\top P\, e\, x^\top \Gamma$, from the adaptive law. Both terms are present, so $\dot{\delta}_i$ can have either sign.

- **p_s and β rows** (constant c_i): $\dot{c}_i = 0$, and the expression simplifies to

$$\dot{\delta}_i = -\frac{\delta_i}{2\,\bar{V}}\, e^\top \tilde{Q} e \;\le\; 0 \qquad (\tilde{Q} \succ 0).$$

In the code: `s = diag(C_lim*P_inv*C_lim')`, `s_dot = 2*diag([-[K_I_dot K_P_dot]; zeros(2,5)]*P_inv*C_lim')`, `delta = sqrt(s*V_bar)` and `delta_dot = (s_dot*V_bar - s*eQe)./(2*sqrt(s*V_bar))`.

Theorem 2 (guaranteed safety with unknown A). Assume (A1)–(A6) and the feasibility condition

$$\text{(F)} \qquad 2\,\delta_i(t) + \frac{2\,\dot{\delta}_i(t)}{\alpha_i} \;<\; y_i^{\max} - y_i^{\min} \quad \text{for all } t \ge 0 \text{ and all } i,$$

together with $x_{\text{ref}}(0)$ inside the shrunken box at $t = 0$. Then for all $t \ge 0$ and every i:

1. the reference model stays in the shrunken box: $y_i^{\min} + \delta_i(t) \le c_i(t) x_{\text{ref}}(t) \le y_i^{\max} - \delta_i(t)$;
2. the **plant stays in the original box**: $y_i^{\min} \le c_i(t) x(t) \le y_i^{\max}$. That is, $u_{\text{bl}} = -\hat{K} x \in [u^{\min}, u^{\max}]$, and p_s and β stay within their limits.

Proof. (1) Let $\tilde{h}_i = c_i x_{\text{ref}} - y_i^{\max} + \delta_i$. Differentiating along the reference model, which is known exactly,

$$\dot{\tilde{h}}_i + \alpha_i \tilde{h}_i = \dot{c}_i x_{\text{ref}} + c_i\big(A_{\text{ref}} x_{\text{ref}} + B_\pi(\pi_0 + \pi) + L_v e\big) + \dot{\delta}_i + \alpha_i \tilde{h}_i = a_i + (H_u \pi)_i.$$

So a_i is the value $\dot{\tilde{h}}_i + \alpha_i \tilde{h}_i$ would have without CBF action ($\pi = 0$), and $(H_u\pi)_i$ is the correction the CBF adds. It remains to show $a_i + (H_u\pi)_i \le 0$, in three steps.

- **The correction.** The CBF uses $\pi = H_u^{-1}\big(\max(0, b) - \max(0, a)\big)$. Since H_u is invertible (A4), $H_u H_u^{-1} = I$, so row by row

  $$(H_u\pi)_i = \max(0, b_i) - \max(0, a_i), \qquad \text{hence} \qquad a_i + (H_u\pi)_i = a_i + \max(0, b_i) - \max(0, a_i).$$

- **Max and min cannot both be active.** By (F), $a_i + b_i = 2\dot{\delta}_i - \alpha_i\big(y_i^{\max} - y_i^{\min} - 2\delta_i\big) < 0$.
- **Check every case.**

Case	Correction $(H_u\pi)_i$	$a_i + (H_u\pi)_i$
$a_i > 0$ (max side active)	$b_i < -a_i < 0$, so $-a_i$	0: exactly cancelled
$a_i \le 0,\ b_i \le 0$ (nothing active)	0	$a_i \le 0$
$a_i \le 0,\ b_i > 0$ (min side active)	b_i	$a_i + b_i < 0$

In every case $a_i + (H_u\pi)_i \le 0$, i.e. $\dot{\tilde{h}}_i + \alpha_i \tilde{h}_i \le 0$. This is the content of Lemma 1, whose hypothesis $a_i + b_i < 0$ is exactly what (F) supplies. The condition is used in the first row, where it rules out the min side being active together with the max side, and in the third row, where it keeps the min-side correction from pushing the max side over. Without it, if $a_i > 0$ and $b_i > 0$ at once, then $(H_u\pi)_i = b_i - a_i$ and $a_i + (H_u\pi)_i = b_i > 0$: the two limits would demand opposite corrections and the max condition would fail.
Fact 2 (with $\kappa(r) = \alpha_i r$) now gives $\tilde{h}_i(t) \le e^{-\alpha_i t}\tilde{h}_i(0) \le 0$. The min side is identical: $\tilde{h}_i^{\min} = y_i^{\min} + \delta_i - c_i x_{\text{ref}}$ satisfies $\dot{\tilde{h}}_i^{\min} + \alpha_i \tilde{h}_i^{\min} = b_i - (H_u\pi)_i \le 0$ by the same three cases. No unknown quantity appears anywhere in this step, which is why it holds exactly.
(2) $c_i x = c_i x_{\text{ref}} + c_i e \le \big(y_i^{\max} - \delta_i\big) + \delta_i = y_i^{\max}$, by (1) and Lemma 3(a). The min side is identical. ∎

Remarks on Theorem 2.

- **Your intuition, made precise.** The shrunken box is enforced *exactly* on the reference model. The plant is only guaranteed to stay within δ_i of the reference model, so it may cross the shrunken bound by up to δ_i, but never the original one.
- **Condition (F).** For p_s and β, $\dot{c}_i = 0$, so $\dot{\delta}_i \le 0$ and (F) reduces to $\delta_i(0) < (y_i^{\max} - y_i^{\min})/2$. For the u_{bl} rows, $\dot{\delta}_i$ can be positive while $\hat{K}$ moves, so (F) should be monitored online. Alternatively, replace $c_i P^{-1} c_i^\top$ by a constant upper bound; Lemma 3(b) bounds $\|\hat{K}(t) - \hat{K}(0)\|_F \le (1 + \sqrt{\mathrm{cond}(\Gamma)})\theta_{\max}$. Then δ_i is non-increasing and (F) only needs to hold at $t = 0$, at the price of more conservatism.
- **The original code omitted $\dot{\delta}_i$.** For the output rows this is harmless because $\dot{\delta}_i \le 0$, which makes the condition conservative. For the input rows it introduces a small, unguarded term. It also omitted $C_{\mathrm{lim}} L_v e$, which is zero there because $L_v = 0$.
- **The total command is not guaranteed.** As in both papers, the bound is imposed on $u_{\mathrm{bl}} = -\hat{K}x$. Theorem 2 says nothing about the total command $u_{\mathrm{bl}} + w$, and it can fail. With $L_v = 0$ and $\gamma = 10^5$, the tube case kept u_{bl} inside its limits while the total rudder reached $1.33\times$ its limit (Table 2). The ACC paper's Theorem 3 argues $|u_{\mathrm{bl}} + w| \le |u_{\mathrm{bl}}|$ when w opposes u_{bl}, but nothing here forces w to do so.
- **Price.** Conservatism: the reference model is held to $y^{\max} - \delta$. The tube shrinks only as $\sqrt{\bar{V}(t)}$, and $\bar{V}$ decreases only by the tracking-error energy. Since the gains need not converge (no persistent excitation), δ does not go to zero. Even with exact convergence, $\bar{V}$ only falls to $\bar{V}_0 - V(0)$, i.e. $(\theta_{\max}^2 - \|\Theta\|_F^2)/\gamma$ for $\Gamma = \gamma I$. Increasing γ is the main way to shrink it.

> **Corollary 2 (no tube on the reference model: the "no tube" case of `Robust_Adaptive_TK.m`).** Run the reference-model CBF of Theorem 2 with $\delta \equiv 0$. Then the reference model stays in the original box, and the plant satisfies
>
> $$y_i^{\min} - |c_i(t)e(t)| \;\le\; c_i(t)x(t) \;\le\; y_i^{\max} + |c_i(t)e(t)|, \qquad |c_i(t)e(t)| \le \sqrt{c_i(t)P^{-1}c_i(t)^\top\, \bar{V}(t)}.$$

Proof. With $\delta \equiv 0$, $a_i + b_i = -\alpha_i(y_i^{\max} - y_i^{\min}) < 0$ always, so step (1) of Theorem 2 gives $y_i^{\min} \le c_i x_{\mathrm{ref}} \le y_i^{\max}$. Then $c_i x = c_i x_{\mathrm{ref}} + c_i e$, and Lemma 3(a) bounds $|c_i e|$. ■

Consequence: the no-tube violation of u_{bl}, p_s and β is capped by the same number that the tube would subtract. Whenever the tube would fit inside the box, this violation is less than the half-width of the box. In practice it is far smaller, because $|c_i e|$ uses only a few percent of the bound. This is why the no-tube runs exceed these limits by only 0.1–0.3%, while their large violations show up in the total command, which neither case bounds.

5. Without shrinking the bounds: a margin on the derivative

> **Proposition 3 (robust CBF on the plant).** Evaluate the CBF on x as in Theorem 1, but add to both a_i and b_i the computable margin
>
> $$\rho_i(t) = \|c_i(t)B\|_2 \sqrt{\lambda_{\max}(\Gamma)\,\bar{V}(t)}\;\|x(t)\|_2.$$
>
> If $2\rho_i(t) < \alpha_i\big(y_i^{\max} - y_i^{\min}\big)$ for all t, then $y_i^{\min} \le c_i(t)x(t) \le y_i^{\max}$ for all $t \ge 0$. The original bounds are not shrunk.

Proof. By Lemma 3(b), $|d_i| \le \|c_i B\|\,\|\Delta K\|_F\,\|x\| \le \rho_i$. The feasibility assumption makes Lemma 1 applicable to $(a + \rho,\ b + \rho)$, so $a_i + \rho_i + (H_u\pi)_i \le 0$. Hence $\dot{h}_i + \alpha_i h_i = a_i + (H_u\pi)_i + d_i \le d_i - \rho_i \le 0$. The min side is identical. ■

For $\Gamma = \gamma I$, $\sqrt{\lambda_{\max}(\Gamma)\bar{V}_0} = \theta_{\max}$ regardless of γ. Unlike the tube, this margin does *not* shrink as adaptation gets faster. Take a representative $\|x\| = 0.3$ during the turn. With $\|c_i B\| = [2.82,\ 4.67,\ 8.41,\ 0.0153]$, the margins are $\rho \approx [1.27,\ 2.10,\ 3.78,\ 0.0069]$, against feasibility limits $\alpha_i y_i^{\max} = [4.19,\ 0.279,\ 12.6,\ 0.035]$. **The rudder row is infeasible.** In addition, the β margin is multiplied by about 65 in H_u^{-1}, which demands roughly 26° of rudder. For this aircraft, the tube is the better place to put the margin.

Remark 4 (some margin is unavoidable). Nagumo's condition is also *necessary*: a set $\{h \le 0\}$ is forward invariant only if $\dot{h} \le 0$ whenever $h = 0$. At a boundary point with $x \neq 0$, the true derivative equals the computable part plus $d_i = c_i B(K^\star - \hat{K})x$. This term is affine in the unknown $K^\star$. Unless $c_i B = 0$, some admissible $K^\star$ (within the $\theta_{\max}$ ball) makes $d_i > 0$. A controller that drives the computable part to exactly zero at the boundary, as the margin-free CBF does, can therefore be pushed out by an admissible plant. To be safe for *every* admissible Θ, the controller needs a margin: either in the state (the tube) or in the derivative (Proposition 3). Both margins scale with $\theta_{\max}$. With no prior bound on Θ, no finite margin suffices, so no guarantee is possible. (This is an informal argument, not a theorem.)

6. Numbers for this aircraft

Table 1 lists the tube width at $t = 0$, $\delta_i(0) = \sqrt{c_i P^{-1} c_i^\top}\,\theta_{\max}/\sqrt{\gamma}$, with $\theta_{\max} = 1.5$ and c_i evaluated at $\hat{K}(0) = K_{\mathrm{lqr}}$. A design is feasible only if $\delta_i(0)$ is below the limit, since the bounds are symmetric.

Design	aileron (lim 3°)	rudder (lim 2°)	p_s (lim 18 dps)	β (lim 0.5°)
Original: $L_v = 0$, $P =$ lyap, $\gamma = 10^4$	1.105	**3.987**	2.614	**0.783**
$L_v = 0$, $P =$ **lyap**, $\gamma = 5 \cdot 10^4$ **(current TK)**	0.494	1.78	1.17	0.350
$L_v = 0$, $P =$ lyap, $\gamma = 10^5$	0.350	1.26	0.827	0.248
Observer-like RM, $\gamma = 10^3$ (earlier TK)	2.850	**6.540**	2.731	**2.601**
Observer-like RM, $\gamma = 10^4$	0.901	**2.068**	0.864	**0.823**
Observer-like RM, $\gamma = 10^5$	0.285	0.654	0.273	0.260
Observer-like RM, $\gamma = 3 \cdot 10^5$	0.165	0.378	0.158	0.150
Observer-like RM, $\gamma = 10^6$	0.090	0.207	0.086	0.082

Table 1. Tube width $\delta_i(0)$ in deg (dps for p_s). Red entries leave an empty shrunken box: the guarantee is impossible.

The original design was infeasible from the start, which is why the clamp to `0.9*Y_max` was needed. A clamp gives a tube that no longer bounds $|c_i e|$, so Theorem 2 no longer applies.

Numerical check of Theorem 2: open-loop reference model (current `Robust_Adaptive_TK.m`)

Setup: the tube design of Section 4 with $L_v = 0$, $P =$ `lyap(A_ref',I)` and $\alpha = \mathrm{diag}(80, 8, 40, 4)$. Uncertain plant $\Theta = 0.5[0\ K_P]$, the same command as before, RK4 with $\Delta t = 10^{-4}$. The first two columns use the settings of the current file.

Check (deg; dps for p_s)	**No tube, $\gamma = 5 \cdot 10^4$**	**Tube, $\gamma = 5 \cdot 10^4$**	**Tube, $\gamma = 10^5$**
Tube width $\delta(0)$ [ail, rud, p_s, β]	0	[0.49, 1.78, 1.17, 0.35]	[0.35, 1.26, 0.83, 0.25]
Guaranteed quantities: peaks [$u_{\mathrm{bl,ail}}$, $u_{\mathrm{bl,rud}}$, p_s, β] vs [3, 2, 18, 0.5]	[2.94, 1.20, 15.38, **0.5015**]	[2.48, 0.30, 15.18, 0.159]	[2.59, 0.70, 15.25, 0.263]
Lemma 3(a): $\max_t \left(\lvert c_i e\rvert - \delta_i\right) \le 0$	n/a	[−0.48, −1.6, −1.1, −0.32]	[−0.34, −1.1, −0.73, −0.22]
(F): minimum margin over t (must be > 0)	—	[6.4, 0.043, 24, 0.021]	[6.8, 0.17, 24, 0.035]
Total command peaks $\lvert u\rvert$ [ail, rud] vs [3, 2] (not covered by Theorem 2)	[2.94, **3.54**]	[2.83, 1.66]	[2.67, **2.66**]

Table 2. Open-loop reference model. The current file, which uses 10 Euler sub-steps instead of RK4, reproduces the first two columns to within 0.04°: total command [2.94, 3.51] without the tube and [2.83, 1.65] with it; β peaks 0.501° and 0.159°.

Theorem 2 holds in both tube runs. The no-tube run crosses the β limit only slightly, as Corollary 2 predicts. Its large violation is in the total rudder command, about 3.5° against 2°. That quantity is outside the theorem. The tube case at $\gamma = 5 \cdot 10^4$ stays inside it in this run, but the tube case at $\gamma = 10^5$ does not (2.66°). The open-loop reference model also brings back the oscillatory adaptation transient in the controls and in v, mainly between 20 and 25 s. Note that the rudder feasibility margin of 0.043 at $\gamma = 5 \cdot 10^4$ is small.

Numerical check of Theorem 2: observer-like reference model

Setup: the tube design of Section 4 (CBF on x_{ref}; $\delta_i(t)$ from $\bar{V}(t)$; $\dot{\delta}_i$ and $C_{\lim} L_v e$ included; observer-like RM with $\nu = 0.1$; $\alpha = \text{diag}(80, 8, 40, 4)$). Uncertain plant $\Theta = 0.5[0\ K_P]$, the same command as before, RK4 with $\Delta t = 10^{-4}$. Each line checks one claim of the theorem.

Check (deg; dps for p_s)	**Tube, $\gamma = 10^5$**	**Tube, $\gamma = 3 \cdot 10^5$**	**No tube, $\gamma = 10^3$**	**No tube, $\gamma = 10^5$**
Tube width $\delta(0)$ [ail, rud, p_s, β]	[0.29, 0.65, 0.27, 0.26]	[0.17, 0.38, 0.16, 0.15]	0	0
Claim 1: $\max_t \left(\lvert c_i x_{\text{ref}} \rvert - (y_i^{\max} - \delta_i)\right) \le 0$	holds (β row on boundary)	holds	holds	holds
Lemma 3(a): $\max_t \left(\lvert c_i e \rvert - \delta_i\right) \le 0$	[−0.26, −0.62, −0.26, −0.25]	[−0.15, −0.36, −0.15, −0.15]	n/a (actual $\lvert c_i e \rvert$ up to [0.05, 0.23, 0.07, 0.027])	n/a (actual $\lvert c_i e \rvert$ up to [0.02, 0.06, 0.02, 0.005])
Claim 2: plant peaks [$u_{\text{bl,ail}}$, $u_{\text{bl,rud}}$, p_s, β] vs [3, 2, 18, 0.5]	[2.55, 0.74, 15.33, **0.248**]	[2.52, 0.88, 15.37, **0.354**]	[2.61, 1.22, 15.44, **0.509**]	[2.65, 1.07, 15.39, **0.5002**]
Total command peaks $\lvert u \rvert$ [ail, rud] vs [3, 2]	[2.55, 1.19]	[2.52, 1.21]	[2.61, 1.22]	[2.65, 0.86]
(F): minimum margin over t (must be > 0)	[7.5, 0.37, 24.8, 0.034]	[7.9, 0.45, 24.9, 0.049]	—	—
Lemma 3: $\max_t(V - \bar{V}) \le 0$; $\bar{V}(40)/\bar{V}(0)$	$-4 \cdot 10^{-6}$; 0.93	$-1 \cdot 10^{-6}$; 0.95	$-4 \cdot 10^{-4}$; 0.97	$-4 \cdot 10^{-6}$; 0.94

Table 3. Simulation check of Theorem 2 with the observer-like reference model. "No tube" means the same reference-model CBF with $\delta \equiv 0$ (Corollary 2). The earlier `Robust_Adaptive_TK.m` (CBF on the plant, Theorem 1, $\gamma = 10^3$, Euler) gave peaks [2.61, 1.21, 15.44, 0.5005] for comparison.

All claims of Theorem 2 hold in both tube runs: the reference model stays in the shrunken box, the tube inequality holds, (F) stays positive, and the plant never leaves the original box. The tube is conservative. The actual $\lvert c_i e \rvert$ stays far below δ_i, and $\bar{V}$ shrinks by only 5–7% over 40 s because the gains do not converge. So β is held at 0.25° or 0.35° instead of 0.5°. Without the tube, the same reference-model design lets the plant cross the β limit (0.509° at $\gamma = 10^3$), which is exactly what your tube was meant to prevent. These runs used RK4. A single Euler step of 10^{-4} s diverges at $\gamma = 10^5$, which is why the code uses sub-steps (Section 7).

7. What this means for the code

- **The current `Robust_Adaptive_TK.m`** runs Theorem 2 (tube) and, for comparison, Corollary 2 (same design with $\delta \equiv 0$). It uses $L_v = 0$, $P =$ `lyap(A_ref',I)`, $\gamma = 5 \cdot 10^4$ and $\theta_{\max} = 1.5$, and prints the three tube checks. Relative to your original tube code, the changes are:
  1. no clamp on δ;
  2. γ large enough that (F) holds;
  3. $\dot{\delta}_i$ and $C_{\lim} L_v e$ added to a and b;
  4. $x(0) = x_{\text{ref}}(0)$;
  5. (F) monitored online;
  6. Euler sub-steps.

Expect conservative behaviour: β is held near $0.5^\circ - \delta_\beta$.

- **If you change settings,** keep (F) positive and check the total command in simulation. The observer-like reference model (the commented line in the file) removes the oscillatory transient, but then needs $\gamma \approx 10^5$ for a feasible tube.
- **If small, bounded violations are acceptable,** the earlier design (CBF on the plant, Theorem 1, observer-like RM, $\gamma = 10^3$) tracks better. Its violations are of order $\sup |d_i|/\alpha_i$ and vanish as adaptation settles.
- **Discretization.** The proofs are in continuous time. A single Euler step of 10^{-4} s diverges at $\gamma = 10^5$. The code therefore takes 10 Euler sub-steps per 10^{-4} s step and holds the CBF output (v, w) over each step, as a sampled-data controller would. The proofs do not cover this discretization, but the runs match RK4 to within 0.04°.
- **Scope of these proofs:** relative-degree-one rows; H_u invertible; the input bound is on u_{bl}, not on the total command $u_{\text{bl}} + w$.

References

[Ames et al. 2017] A. D. Ames, X. Xu, J. W. Grizzle, P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Trans. Automatic Control*, 62(8):3861–3876, 2017.

[Blanchini 1999] F. Blanchini, "Set invariance in control," *Automatica*, 35(11):1747–1767, 1999.

[Khalil 2002] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002.

[ACC 2025] E. Lavretsky, M. Menner, "Servo-controllers for linear time-invariant systems with operational constraints," *American Control Conference*, pp. 4909–4916, 2025.

[Lavretsky & Wise 2024] E. Lavretsky, K. A. Wise, *Robust and Adaptive Control: With Aerospace Applications*, 2nd ed., Springer, 2024.

[SciTech 2026] M. Menner, E. Lavretsky, "Robust linear design for flight control systems with operational constraints," *AIAA SciTech Forum*, AIAA 2026-1742, 2026.

[Nagumo 1942] M. Nagumo, "Über die Lage der Integralkurven gewöhnlicher Differentialgleichungen," *Proc. Physico-Mathematical Society of Japan*, 24:551–559, 1942.

[Nguyen & Sreenath 2016] Q. Nguyen, K. Sreenath, "Exponential control barrier functions for enforcing high relative-degree safety-critical constraints," *American Control Conference*, pp. 322–328, 2016.